To **prove the formula for $\boldsymbol{\theta}$** (the parameter vector) in **Linear Regression**, we need to use **Ordinary Least Squares (OLS)** and derive the **closed-form solution**.

🧠 Goal:

Minimize the **cost function** (Mean Squared Error) for linear regression:

$$J(\boldsymbol{\theta}) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2$$

Where:

- m: number of training examples
- $h_\theta(x^{(i)}) = \boldsymbol{\theta}^T \mathbf{x}^{(i)}$ is the hypothesis
- $\mathbf{x}^{(i)}$: feature vector (including 1 for bias term)
- $y^{(i)}$: target output

🧮 Step-by-step Derivation Using Matrix Notation

Let:

- $\mathbf{X} \in \mathrm{R}^{m \times n}$: design matrix (rows = examples, columns = features)
- $\mathbf{y} \in \mathrm{R}^{m \times 1}$: vector of outputs
- $\boldsymbol{\theta} \in \mathrm{R}^{n \times 1}$: parameter vector

Step 1: Cost Function in Matrix Form

$$J(\boldsymbol{\theta}) = \frac{1}{2m} (\mathbf{X}\boldsymbol{\theta} - \mathbf{y})^T (\mathbf{X}\boldsymbol{\theta} - \mathbf{y})$$

Step 2: Take the Derivative w.r.t. $\boldsymbol{\theta}$

We compute the gradient:

$$\nabla_\theta J(\boldsymbol{\theta}) = \frac{1}{2m} \cdot 2\mathbf{X}^T (\mathbf{X}\boldsymbol{\theta} - \mathbf{y}) = \frac{1}{m} \mathbf{X}^T (\mathbf{X}\boldsymbol{\theta} - \mathbf{y})$$

Step 3: Set the Gradient to Zero to Minimize Cost

$$\mathbf{X}^T (\mathbf{X}\boldsymbol{\theta} - \mathbf{y}) = 0$$

$$\Rightarrow \mathbf{X}^T\mathbf{X}\boldsymbol{\theta} = \mathbf{X}^T\mathbf{y}$$

Step 4: Solve for $\boldsymbol{\theta}$

$$\boldsymbol{\theta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

✅ Final Formula

$$\boxed{\boldsymbol{\theta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}}$$

This is called the **Normal Equation**.